

Multi-Tenant School Administration Management SaaS Platform

School Portal

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Academic Year: 2025–26

MCA Major Project — Qollabb Submission

Problem Statement

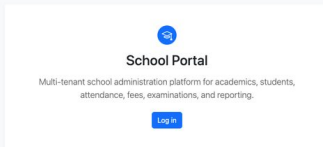
- Schools often rely on paper registers, spreadsheets, and disconnected tools.
- The same student data is entered repeatedly, causing duplication and inconsistency.
- Manual attendance, fee, and examination work is slow and error-prone.
- Reporting requires collecting data from many sources with little automation.
- Weak access control and limited audit evidence increase security risk.
- Multi-school organizations lack a single secure, tenant-aware platform.

Objectives and Scope

- Build a multi-tenant SaaS platform for school administration workflows.
- Enforce tenant isolation and role-based access for four fixed roles.
- Deliver 15 modules: auth, schools, users, academics, students, attendance, fees, exams, reports, dashboards, logs, backups.
- Support Super Admin, School Admin, Teacher, and Accountant responsibilities.
- Demonstrate engineering practice: layered architecture, services, policies, tests, deployment.
- Out of scope: parent/student portals, mobile app, live payment gateway, custom roles.

Existing vs Proposed System

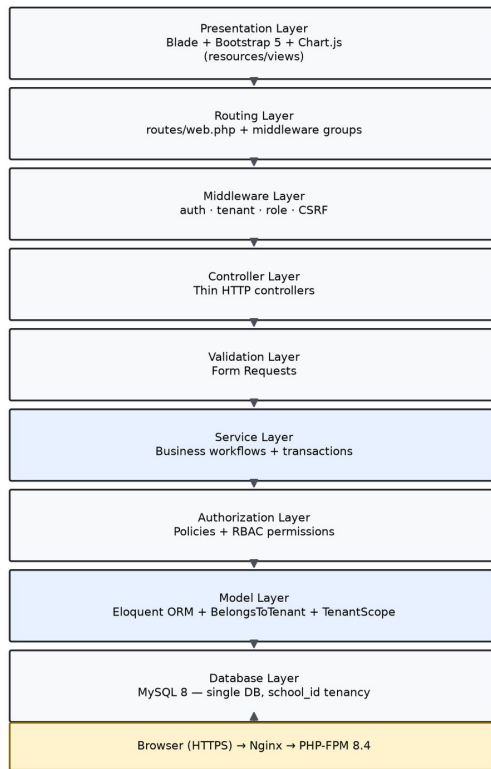
- Existing: manual registers, spreadsheets, separate fee and exam files.
- Existing: slow retrieval, human calculation errors, weak accountability.
- Proposed: School Portal — one secure web application for all core workflows.
- Proposed: structured academic setup, student records, and enrollment tracking.
- Proposed: attendance entry, sandbox fee collection, examination and report cards.
- Proposed: dashboards, operational reports, activity log, audit trail, backup evidence.



Architecture and Native Multi-Tenancy

- Layered monolith: Blade UI → routes/middleware → controllers → services → models → MySQL.
- Native Laravel tenancy using `school_id` — no external tenancy package.
- `TenantContext` middleware, global scopes, `BelongsToTenant`, policies, and services.
- Single application and shared schema; each school is a logically isolated tenant.
- Super Admin (`school_id` NULL) uses authorized platform paths only.
- Stack: Laravel 13, PHP 8.4, MySQL 8, Bootstrap 5, Git/GitHub.

**Figure 4.1 — Layered System Architecture
School Portal (Laravel 13 Monolith)**



Database and ER Design

- MySQL 8 relational database with shared-schema multi-tenant design.
- 28 tables covering platform, tenant, academic, student, attendance, fee, exam, and operations data.
- Tenant-owned records linked through `school_id` with indexes and foreign keys.
- `restrictOnDelete()` preserves academic, financial, and audit history.
- Normalization reduces duplication across students, enrollments, and transactions.
- ER design supports reporting, RBAC, and tenant isolation testing.

Figure 4.2 — Core Entity Relationship Diagram
28 Tables (logical groups)

Platform



Academic / Students



Attendance / Fees / Exams



System

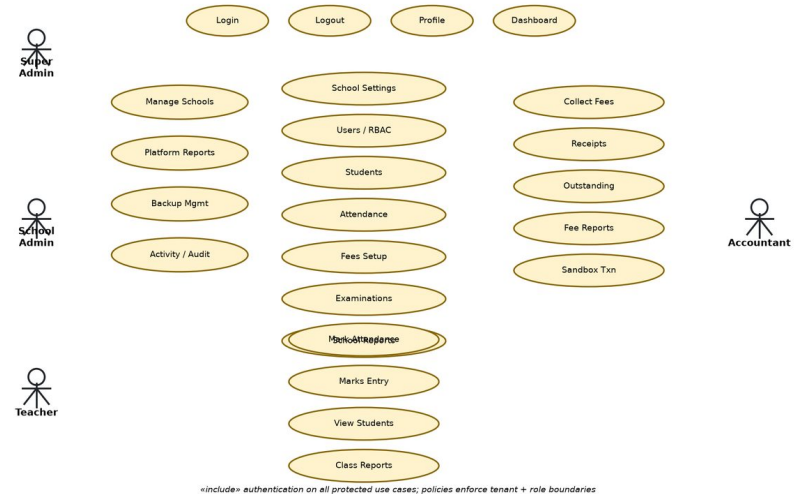


Tenant-owned tables include `school_id` · Platform tables: `roles`, `permissions`, `role_permissions`

Roles and Authorization (RBAC)

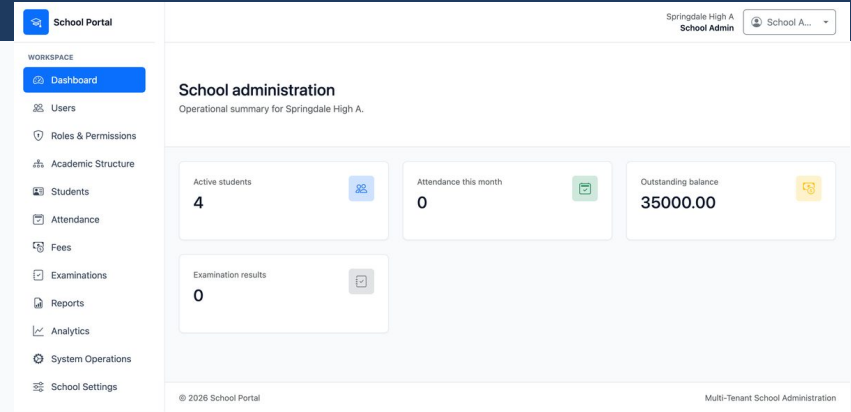
- Super Admin: platform dashboard, school management, platform reports, backups.
- School Admin: own-school settings, users, academics, students, attendance, fees, exams, reports, logs.
- Teacher: assigned attendance, marks entry, and scoped report-card access.
- Accountant: fee collection, receipts, outstanding balances, financial reports.
- Authorization via Laravel Policies, Gates, middleware, and Form Request validation.
- Fixed four-role model keeps permissions testable and secure.

Figure 3.3 — Use Case Diagram
School Portal (Four Canonical Roles)



Core Academic and Student Modules

- Academic structure: years, terms, classes, sections, subjects, teacher profiles.
- Student registration, profile management, photo handling, and enrollment.
- School Admin dashboard for operational visibility and navigation.
- Teacher dashboard for assigned class and attendance workflows.
- Attendance entry with date/class/section filtering and summary views.
- Module design keeps controllers thin and business rules in services.



Attendance, Fees, and Examinations

- Attendance: daily entry, review, and reporting by class and date.
- Fees: structures, collection, receipts, outstanding balances (sandbox transactions only).
- Examinations: exam setup, marks entry, grade scales, result processing.
- Report cards: printable academic outputs with assignment-scoped teacher access.
- Workflows use service-layer transactions and policy checks.
- Activity and audit evidence recorded for sensitive operations.

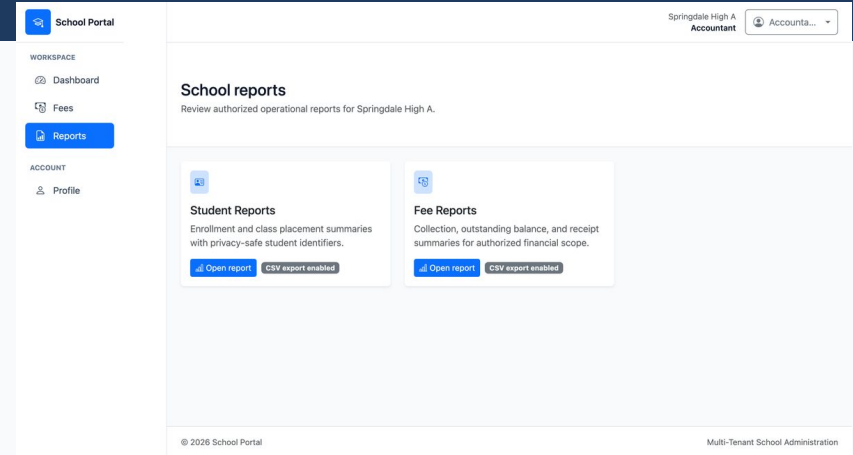
The screenshot displays the 'School Portal' interface for an 'Accountant' user at 'Springdale High A'. The left sidebar shows navigation options under 'WORKSPACE' (Dashboard, Fees, Reports) and 'ACCOUNT' (Profile). The 'Fees' option is selected. The main content area is titled 'Fee Collection' with a subtitle 'Collect payments against open Student Fee assignments in the active school.' Below this, there are tabs for 'Payment History', 'Outstanding Balances', 'Sandbox Transactions', and 'Fee Collection' (which is active). A search bar is present with the placeholder 'Admission number or student name' and a 'Search' button. The main table lists fee assignments with columns: Student, Fee Category, Class, Balance, State, and Actions. The table contains three rows of data, each with a 'Collect' button in the Actions column.

Student	Fee Category	Class	Balance	State	Actions
SHA-001 Aarav Sharma	Tuition	Class 8	15,000.00	Pending	Collect
SHA-002 Isha Patel	Tuition	Class 8	14,000.00	Pending	Collect
SHA-001 Aarav Sharma	Transport	Class 8	6,000.00	Pending	Collect

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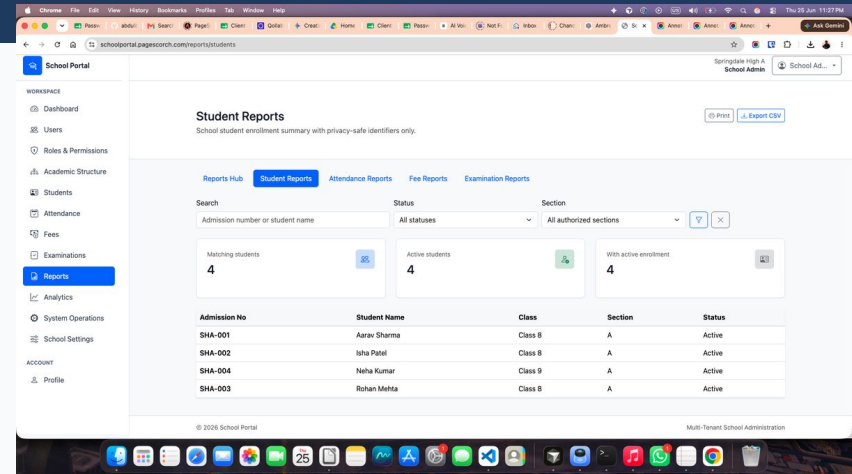
Reports, Analytics, and System Operations

- Reports hub: Student, Attendance, Fee, Examination, School, and User reports.
- Role-specific dashboards with Chart.js summaries where applicable.
- Activity Log: user action evidence for operational monitoring.
- Audit Trail: data-change history for accountability.
- Backup Management: Super Admin backup operation records and export evidence.
- System Operations separated from the operational Reports hub by design.



Security and Tenant Isolation

- Authentication, CSRF protection, session security, and server-side validation.
- Policies and Gates enforce module-level and record-level authorization.
- Tenant isolation via `school_id` scopes — never trust client-supplied tenant keys.
- Private storage for student photos; controlled access to sensitive records.
- Automated tenant-isolation tests plus two-browser SHA/SHB smoke validation.
- Cross-tenant data exposure treated as a critical defect.



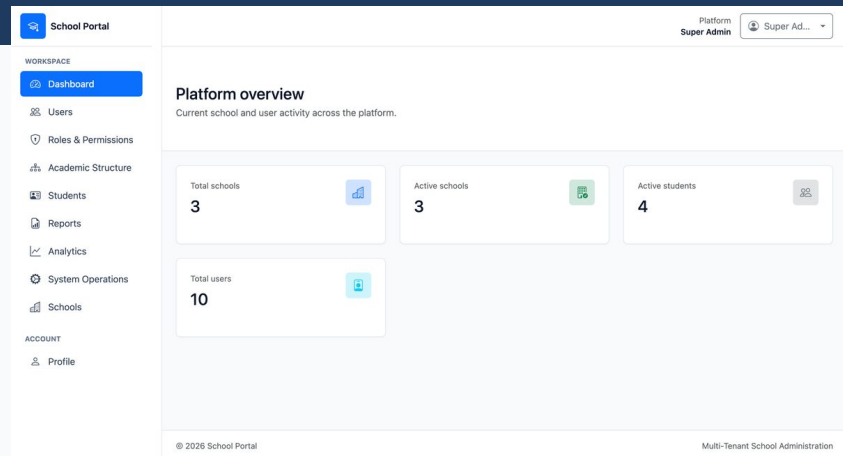
Testing and Validation Results

- 394 automated tests with 3,312 assertions in the Laravel test suite.
- Feature tests for workflows; policy tests for RBAC denials.
- Tenant-isolation, security, and rollback scenarios covered.
- Manual smoke testing across four roles on the deployed application.
- Pint, migrations, seeders, and release gates used during development.
- System considered ready for MCA report evidence and live demonstration.

```
{"tool":"phpunit","result":"passed","tests":394,"passed":394,"assertions":3312,"duration_ms":36928}
```

Cloud Deployment and Live Demo

- Deployed cloud-hosted School Portal for evaluator access.
- Live URL: schoolportal.pagescorch.com (HTTPS).
- Demo schools seeded for multi-tenant demonstration (SHA, SHB, SHC).
- GitHub repository available for source and documentation review.
- Installation, deployment, and smoke-test guides included in project docs.
- Supports Qollabb live-demo and viva evaluation requirements.



Achievements, Limitations, and Future Work

- Achieved: full MCA scope — 15 modules, native multi-tenancy, RBAC, reports, ops evidence.
- Achieved: documented architecture, testing strategy, and deployment package.
- Limitation: sandbox fee workflow only — no production payment gateway.
- Limitation: no parent/student portals, mobile app, SMS, or custom roles.
- Future: parent portal, notifications, live payments, richer analytics, API integrations.
- Future: performance tuning and extended operational monitoring as schools scale.

Conclusion and Thank You

- School Portal centralizes multi-school administration in one secure SaaS platform.
- Native school_id tenancy, RBAC, and layered architecture support maintainability.
- Core academic, attendance, fee, examination, and reporting workflows are implemented.
- 394 tests and live deployment provide strong MCA submission evidence.
- Live demo: <https://schoolportal.pagescorch.com>
- Source code: <https://github.com/abdulbaquee/multi-tenant-school-saas>
- Thank you — questions welcome.